A. Authenticity of the Submitted Work

- The work submitted for this project is our own. Where we used material written by others, it is cited.
- Every figure reported in the IEEE paper comes from a run that actually took place. The runs are recorded in Cache_Final_Project_Experiment_Logs and Cache_Final_Chatbot_Response _Evaluation
- Eight Colab runs of the chatbot were performed, with each run running the chunking and prompt parts for time efficiency. No run was discarded, and no figure was selected from among several runs.
- The researchers evaluated the chatbot through the provided precision@3, recall@3, mrr@3 formulas. It's also used to evaluate the top 3 retrieved chunks for the prompt questions relevant to the source document.

B. Disclosure of Generative AI Use

Generative AI tools were used in this project. The table below records where, and is completed individually in section C.

Where AI was used	What it did	What we did
Code	Drafted and debugged parts of the notebook	Specified the design, ran the code, checked the output against the data, and can explain each section
Writing	Drafted and edited parts of the IEEE report and the documentation	Supplied the findings and numbers, checked every figure against the measured output, and corrected claims that overstated the results
Analysis	Drafted the time log code code to help estimate the average time the chatbot	Logged indexing and total time execution for CPU and GPU runs

	responds for each question	
Not used for	No reported measurement was produced or estimated by an AI tool. Every number comes from an executed run, and the answer scores were assigned by the researchers.	

We understand that disclosing the use of these tools is required, and that the responsibility for the accuracy of everything submitted remains ours.

C. Individual Declaration

Each member completes one block. Describing their own use of generative AI in their own words, then sign. Each member completes and signs themselves, a single signature does not cover the group.

Member: Hanna Alexa Laylo

Components I own and can defend: Evaluated the chatbot's generated responses to determine whether they accurately answered the question based on the source document; calculated Precision@3, Recall@3, and MRR@3 for each question and computed the average scores across all 12 questions; drafted the main body of the IEEE report organized and formatted the excel experiment results for clearer presentation; manually took out the chatbot responses and putting them into the excel spread sheet along with their corresponding questions.

My use of generative AI: Generative AI was used as a supplementary tool to improve the readability, grammar and formality of my initial drafts and written content; It was not used to write the report on my behalf, as original ideas, content, and drafts were prepared by me.

I declare that the above is accurate and complete.

Signature:  Date: 11/11/2026

ITC C508-402I (Group Name: Cache) - Final Project - P-F1 (Group Final Project)

Member: Perlin Precious Sasil

Components I own and can defend: I had Maintained the experiment data curation by organizing and logging the raw outputs into the Excel and conducted the needed experimental investigation across the chatbot runs and authored the initial draft for Section II which is the Related Work / Background, Section III which is the System Architecture / Data Curation, and Section VIII which is the Conclusion / Future Work of the IEEE report.

My use of generative AI: To assist with sentence restructuring, refining technical phrasing, and enhancing readability across my assigned report sections. All underlying data curation, experimental testing, and core section contents were performed and validated manually by me.

I declare that the above is accurate and complete.

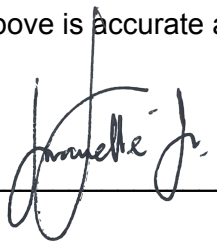
Signature:  Date: 11/11/2026

Member: Nanelle Tuguinay

Components I own and can defend: The chunking experimentation values and prompt questions gathered from our last activity. The manual double checking of average time the chatbot takes to respond per question.

My use of generative AI: Drafting the time logging code lines in the code to assist with time logging of each run, assisting time estimates of the chatbot's response time per question. And assistance in debugging the application of time logging to the chatbot.

I declare that the above is accurate and complete.

Signature:  Date: 11/11/2026